Which retrieval settings find the true value?

Truth coverage and graded relevance of the documents medsim retrieves (Europe PMC + LitSense), on the 141 set A questions of the retrieval benchmark · judge anthropic/claude-haiku-5.5 · 37 scored configurations and controls

> **Result.** The best configuration is **llm_select20+epmc25+q_synonym** (an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion): strict truth coverage 70.2% against the baseline's 60.3% (+9.9 (+3.5 to +17.0) points, paired over the same questions), nDCG@8 0.750 against 0.691, and precision@8 49.6% against 41.3%. On fresh queries: query set fresh1: strict +7.1 points, nDCG +0.054; query set fresh2: strict +12.1 points, nDCG +0.058.

What was measured

- **Strict truth coverage** (decides): a document gives a value or range for this variable in patients with this or a closely related condition, and the hidden true value lies inside it or within about 15%. **Category coverage** (shown): a document's value, range, or level, a reference range for comparable patients included, puts the true value in the same low / normal / high category.
- **Graded relevance**: each document 0 (not about the variable or the condition), 1 (about them, no usable value), or 2 (a value for the variable in comparable patients). **nDCG@8** is against a perfect list of eight grade-2 documents, so it rewards better documents and a better order (it breaks ties); **precision@8** is the share of grade-2 documents.
- One judge call per question grades all eight documents and gives both coverage verdicts. Every positive verdict needs a verbatim quote that code finds in the named document; an unverified grade 2 counts as 1, and unverified coverage as not covered.
- Each configuration retrieves for the same **frozen search query** per question (medsim's Stages A and B once, with the stored runs' model deepseek/deepseek-v4-flash-0731), and no answer is written. Differences between configurations therefore come from the retrieval settings. Eight documents per question throughout. Search results are cached, so a configuration that issues the same requests gets the same results.

Can the metrics be trusted?

Check	Strict coverage	Category	nDCG@8	Reading
Shuffled control: each question judged on another question's documents	0.0%	0.7%	0.034	Near zero: the judge reads the documents rather than answering from the question.
The baseline's documents judged twice	60.3% / 61.7%	83.7% / 85.8%	0.691 / 0.680	The judge's own noise: 1.4 points of strict coverage.
Stored benchmark runs 2–6 of the baseline method (fresh queries, full pipeline)	61.7% ± 3.4	–	0.675 ± 0.019	Run-to-run spread (mean ± SD) when the query is written anew; the frozen-query baseline (60.3%) lies inside it.
Stored run 2 of the original retrieval (`current`)	40.4%	70.2%	0.593	About 20 points below changes 1–4: the metrics separate a known difference.

Single changes from the baseline

The baseline is medsim's default retrieval (changes 1–4: value-first ranking in one list, 50 Europe PMC candidates, animal and age filter, article-body search with excerpts, value-based broadening). Each row changes one setting. Differences are paired over the same 141 questions, with 95% bootstrap intervals; green intervals exclude zero.

Configuration	Strict coverage	Δ strict vs baseline (95% CI)	Category coverage	nDCG@8	Δ nDCG vs baseline	Precision@8	n
baseline baseline: changes 1-4	60.3%		83.7%	0.691		41.3%	141
llm_select40 an LLM picks the 8 documents from the best 40	68.8%	+8.5 (+2.1 to +15.6)	89.4%	0.746	+0.055 (+0.030 to +0.078)	50.8%	141
llm_select20 an LLM picks the 8 documents from the best 20	67.4%	+7.1 (+0.7 to +14.2)	89.4%	0.749	+0.058 (+0.041 to +0.076)	49.6%	141
epmc25 25 Europe PMC candidates (baseline 50)	66.0%	+5.7 (-1.4 to +12.8)	85.1%	0.690	-0.001 (-0.020 to +0.017)	42.6%	141
q_synonym Europe PMC MeSH synonym expansion	63.8%	+3.5 (-1.4 to +8.5)	84.4%	0.690	-0.001 (-0.015 to +0.013)	42.3%	141
epmc_only Europe PMC only	63.1%	+2.8 (-2.8 to +8.5)	85.8%	0.665	-0.026 (-0.048 to -0.005)	40.5%	141
ls_sentences LitSense sentences instead of passages	63.1%	+2.8 (-4.3 to +9.9)	86.5%	0.659	-0.032 (-0.053 to -0.013)	39.3%	141
q_natural LitSense natural-language queries	62.4%	+2.1 (-4.3 to +8.5)	83.7%	0.684	-0.007 (-0.029 to +0.014)	41.3%	141
round_robin round-robin merge across sources instead of one list	62.4%	+2.1 (-4.3 to +8.5)	83.7%	0.667	-0.024 (-0.044 to -0.006)	39.6%	141
ls100 100 LitSense results	61.7%	+1.4 (-5.0 to +7.8)	86.5%	0.681	-0.010 (-0.028 to +0.007)	40.9%	141
epmc100 100 Europe PMC candidates	61.7%	+1.4 (-5.0 to +7.8)	85.8%	0.679	-0.012 (-0.028 to +0.004)	40.5%	141
fulltext20 20 open-access full texts fetched for excerpts (baseline 10)	61.0%	+0.7 (-5.7 to +7.1)	87.9%	0.700	+0.009 (-0.008 to +0.026)	44.7%	141
ls60 60 LitSense results (baseline 30)	61.0%	+0.7 (-5.7 to +7.1)	86.5%	0.681	-0.010 (-0.025 to +0.005)	41.2%	141
ls_no_rerank no LitSense-side rerank	61.0%	+0.7 (-5.7 to +7.1)	87.2%	0.677	-0.014 (-0.033 to +0.004)	40.5%	141
chars3000 3,000 characters per document (baseline 1,500)	60.3%	+0.0 (-6.4 to +6.4)	86.5%	0.677	-0.013 (-0.035 to +0.006)	41.7%	141
no_value_first no value-first ranking	59.6%	-0.7 (-7.1 to +5.7)	84.4%	0.683	-0.008 (-0.026 to +0.009)	41.5%	141
ladder_v1 the original broadening ladder (v1)	59.6%	-0.7 (-7.8 to +6.4)	85.8%	0.661	-0.029 (-0.050 to -0.010)	37.8%	141
q_gpt6luna openai/gpt-6-luna writes the search query	58.9%	-1.4 (-10.6 to +7.1)	87.9%	0.704	+0.014 (-0.016 to +0.040)	44.0%	141
no_population_filter no animal and age-group filter	56.7%	-3.5 (-9.2 to +2.1)	85.8%	0.682	-0.008 (-0.026 to +0.008)	40.7%	141
no_excerpts no full-text excerpts (abstracts and passages only)	39.0%	-21.3 (-29.8 to -12.8)	70.9%	0.597	-0.094 (-0.122 to -0.068)	29.7%	141
no_lexical_rerank no lexical rerank within each source	34.8%	-25.5 (-34.0 to -17.0)	66.0%	0.554	-0.136 (-0.164 to -0.110)	20.3%	141
litsense_only LitSense only	29.8%	-30.5 (-40.4 to -21.3)	61.7%	0.576	-0.115 (-0.144 to -0.087)	23.8%	141

With 21 changes tested at 95%, about one could look better or worse by chance alone; treat single changes whose interval only just excludes zero with care.

Combinations

The changes with the largest positive estimates, added to the best single change.

Configuration	Strict coverage	Δ strict vs baseline (95% CI)	Category coverage	nDCG@8	Δ nDCG vs baseline	Precision@8	n
baseline baseline: changes 1-4	60.3%		83.7%	0.691		41.3%	141
llm_select20+epmc25+q_synonym **an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion**	**70.2%**	**+9.9 (+3.5 to +17.0)**	**87.9%**	**0.750**	**+0.059 (+0.042 to +0.076)**	**49.6%**	**141**
llm_select20+q_synonym an LLM picks the 8 documents from the best 20; Europe PMC MeSH synonym expansion	68.8%	+8.5 (+2.8 to +14.9)	90.8%	0.743	+0.052 (+0.035 to +0.070)	48.8%	141
llm_select20+epmc25 an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50)	68.8%	+8.5 (+1.4 to +14.9)	87.9%	0.737	+0.046 (+0.027 to +0.065)	48.3%	141

Confirmation on fresh queries

The baseline and the best configuration again, each on two new sets of queries written by the same query builder: the gain should hold when the queries change. Here each Δ compares with the baseline on the same query set.

Configuration	Strict coverage	Δ strict vs baseline (95% CI)	Category coverage	nDCG@8	Δ nDCG vs baseline	Precision@8	n
baseline@fresh1 baseline: changes 1-4 (queries: fresh1)	62.4%		87.9%	0.681		40.7%	141
llm_select20+epmc25+q_synonym@fresh1 **an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion (queries: fresh1)**	**69.5%**	**+7.1 (-0.7 to +14.9)**	**89.4%**	**0.735**	**+0.054 (+0.036 to +0.074)**	**47.1%**	**141**
baseline@fresh2 baseline: changes 1-4 (queries: fresh2)	57.4%		85.1%	0.682		41.7%	141
llm_select20+epmc25+q_synonym@fresh2 **an LLM picks the 8 documents from the best 20; 25 Europe PMC candidates (baseline 50); Europe PMC MeSH synonym expansion (queries: fresh2)**	**69.5%**	**+12.1 (+4.3 to +19.9)**	**92.9%**	**0.740**	**+0.058 (+0.039 to +0.078)**	**47.5%**	**141**

Cost and limits

- Cost of this work from the stored usage records: $9.11 (judge $7.00, LLM selection during retrieval $1.79, query writing $0.32). Europe PMC and LitSense are free.
- One judge, an LLM, without human labels: absolute levels may be off; the comparisons between configurations, made by the same judge on the same questions, are the point.
- 141 questions: differences of a few points are inside the noise.
- Europe PMC answered HTTP 503 during the first retrieval of the two fresh-query baseline runs (129 and 64 of 141 questions lost it). Those questions were retrieved again once it recovered and judged again; the tables use the second attempt. Every other run had no failed source.
- LLM selection adds one call per question to medsim's retrieval (about $0.0016 with deepseek/deepseek-v4-flash-0731, and its latency). It is not medsim's default yet.
- The best configuration was chosen on these questions; the fresh-query runs re-test it, but on the same 141 questions, not on new ones.